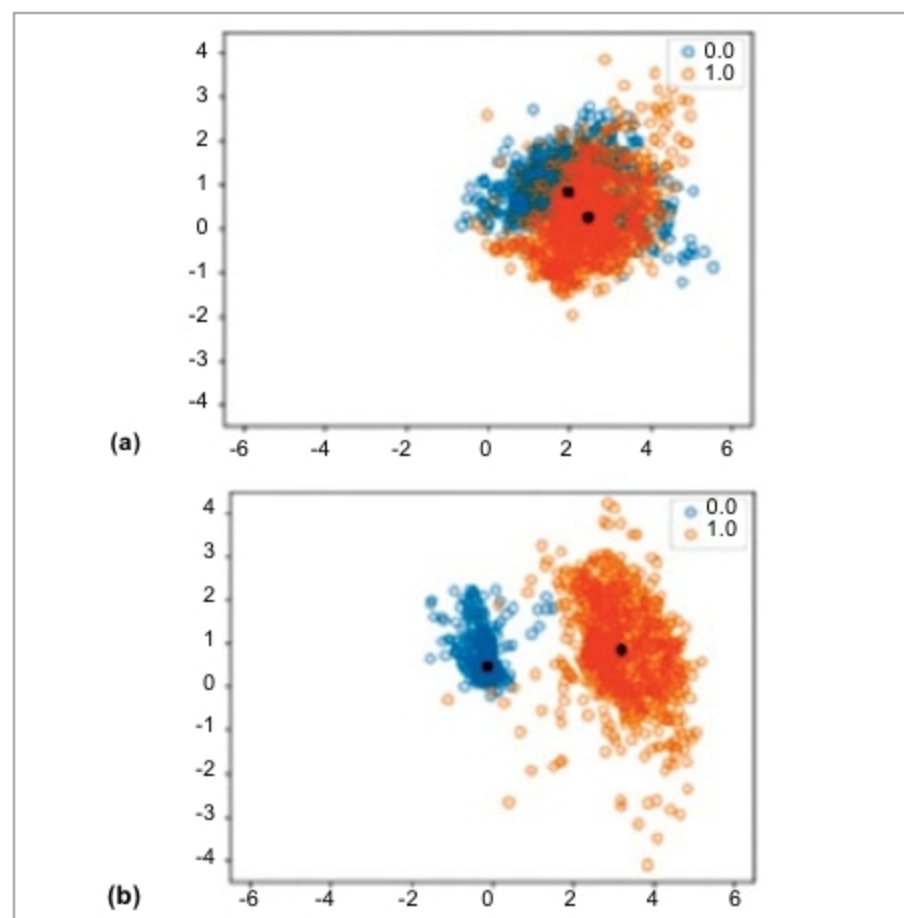


AI system that continually learns

Artificial intelligence (AI) systems are driving technology advancements in many commercial and government applications such as speech recognition and autonomous robotics. However, current systems are not able to handle new scenarios that these are not trained on. AI systems today can repeatedly make the same mistakes. Even with retraining, the systems are prone to catastrophic forgetting when a new item disrupts previously-learned knowledge.



Overlapping tasks in latent space (blue/orange) makes Lifelong RL training prone to forgetting (a), while our generative memory and task encoding separate the latent space to ensure efficient memory use to avoid forgetting (b) (Credit: SRI International)

To address these limitations, SRI International is developing a next-generation AI system that can learn continuously and apply that learning to become better and more reliable at performing new tasks. The work is being done under Defense Advanced Research Projects Agency's (DARPA's) Lifelong Learning Machines (L2M) programme.

SRI researchers are training AI agents using real-time strategy games such as StarCraft II. Using deep reinforcement learning methods, AI agents are trained with surprises injected into the game (for example, terrain and unit capability change). Using this method, key metrics for life-long learning metrics such as adaptation, robustness and safety can be measured.

3D desktop prints 4D multi-metal objects

4D printing can now create complex 3D geometries that react to environmental stimuli, opening new design opportunities in materials science. A vast majority of 4D printing

approaches use polymer materials, which limit the operational temperature during the process of engineering. In a recent study, Xiaolong Chen and co-workers at Dyson School of Design and Engineering, Department of Earth Science and Engineering and Department of Materials at Imperial College of London, UK, have developed a new multi-metal electrochemical 3D printer. The device can construct bimetallic geometries by selectively depositing different metals with temperature-responsive behaviour programmed into the printed structure.

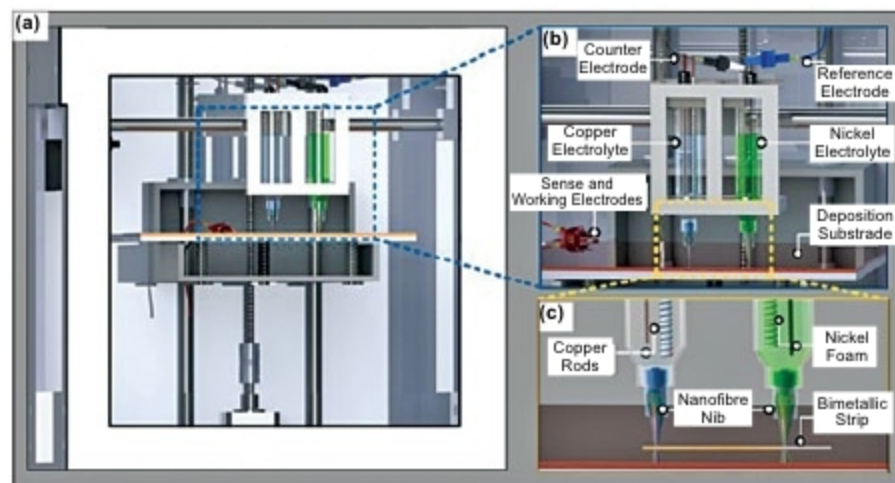


Illustration of low-cost electrochemical multi-metal 3D printer—front view (a), print head setup (b) and detailed view highlighting deposition nozzles and deposited bimetallic strip (c) (Credit: Scientific Reports)

In the study, scientists demonstrated a meniscus confined electrochemical 3D printing approach using a multi-print head design and nickel and copper materials as examples—the ability can be transferred to other deposition solutions.

Additive manufacturing, popularly known as 3D printing, can fabricate complex 3D architectures by sequentially joining materials layer-by-layer. Flexibility of this technique has found applications in aerospace, automotive, medical and energy industries. At first, applications of additive manufacturing focused on the use of polymers to ease consolidation, either via photo-polymerisation (stereolithography) or thermal processes (such as fused deposition modelling). However, increased uptake of metal-based additive manufacturing has now transitioned the technology from a prototyping tool to engineering end products.

Samsung TVs could soon be completely wireless

Last year, Samsung Electronics filed a patent with World Intellectual Property Office (WIPO) for a wireless power transceiver and display apparatus. The patent describes a system in which there will be an electromagnetic bar mounted in between the TV and the wall. It will draw power from a base station that would also double up as a soundbar—that is where the TV will get its power from. This would result in a TV that does not require any cables or wires to keep itself powered. Although, it is to be seen how a set-top box or another device would be attached to it.